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Research and Applications

irAE.AI: AI-powered exploration of real-world immune-related adverse events

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Abstract

Objectives: Immune checkpoint inhibitors (ICIs) have revolutionized cancer treatment, but their clinical benefit is limited by the onset of immune-related adverse events (irAEs). Real-world pharmacovigilance data, such as the FDA's Adverse Event Reporting System (FAERS) database, offer the scale needed to better characterize and understand these toxicities. However, extracting cohorts and normalizing complex fields requires expertise in data preprocessing, technical terminology harmonization, and programming.

Materials and Methods: We curated an oncology-specific FAERS dataset of ICI-treated cases (2012Q4-2025Q3), standardized tumor, drug, and adverse event fields, deduplicated reports, and generated a flat-file resource ($N = 71\,175$ irAE cases). To broaden accessibility, we developed a large language model (LLM)-powered analytics assistant that translates natural language queries into executable Python code for cohort filtering, visualization, and statistical testing. Our system also leverages retrieval-augmented generation (RAG) to return clinical guideline-grounded responses to questions about irAEs and their management for research purposes.

Results: We benchmarked the system on curated tasks (question intent classification, TableQA, statistics, plotting) using multiple LLMs and report model- and task-specific performance metrics. As a case study, the platform recovered established irAE trends including differences in irAE patterns between anti-CTLA-4 treatment and other immunotherapy regimens and tumor-specific differences in irAE profiles following anti-PD-1 treatment.

Discussion: Our results show that LLM-driven analytics can reliably translate natural-language queries into reproducible analyses, though model choice affects accuracy across tasks.

Conclusion: This platform provides a transparent, user-friendly approach for exploring real-world immunotherapy safety data to support hypothesis generation, candidate biomarker identification, and irAE risk assessment in immuno-oncology.

Key words: artificial intelligence, large language models, immune checkpoint inhibitors, drug-related side effects and adverse reactions, real-world evidence

Lay Summary

Immunotherapy helps the immune system attack cancer, but it can also trigger side effects that resemble autoimmune diseases, called immune-related adverse events (irAEs). Understanding these side effects is important to inform safer cancer treatments, but existing data can be difficult to access and analyze. In this study, we created a user-friendly platform to explore real-world irAE data from the FDA's Adverse Event Reporting System (FAERS), a large database of side effects reported by patients, clinicians, and drug manufacturers. We first built a curated dataset focused specifically on cancer patients treated with immunotherapy, which resulted in a cleaned final dataset of over 70 000 irAE cases. We then developed an artificial intelligence (AI) chatbot that allows users to ask questions in plain language and instantly analyze and

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explore the data. The system can filter and summarize data, create visualizations, and perform statistical tests without requiring programming skills. It can also provide responses informed by current clinical guidelines on irAEs and their management. After testing multiple AI models, we selected the best-performing model to power the tool. Overall, this platform offers a simple and flexible way for researchers and clinicians to explore real-world immunotherapy side effects and generate new hypotheses and insights.

Background and significance

The introduction of immunotherapies has revolutionized cancer treatment, leading to durable responses and improved survival across a wide range of malignancies. Immune checkpoint inhibitors (ICIs) enhance anti-tumor immune responses by targeting inhibitory molecules such as programmed cell death protein 1 (PD-1), programmed death ligand 1 (PD-L1), cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4), and lymphocyte activation gene 3 (LAG-3). However, this immune activation can result in immune-related adverse events (irAEs), a distinct class of toxicities that reflect autoimmune and inflammatory responses.

irAEs can affect nearly any organ system, including most commonly the skin, gastrointestinal tract, endocrine organs, lungs, and liver, but also less frequently the cardiovascular, neurologic, and hematologic systems.[1,2] Severity ranges from mild symptoms to life-threatening complications requiring hospitalization and immune suppression. The development of irAEs often results in treatment interruption or permanent suspension of ICIs, which can compromise anti-tumor effects.[1–5] A recent review harmonizing data from over 200 ICI studies (including both clinical trial and real-world data) estimated that about 40% of ICI-treated patients developed an irAE, with approximately 20% experiencing grade ≥3 toxicities.[6] As the indications for ICIs continue to expand, understanding the real-world spectrum and determinants of irAEs is a critical priority in oncology research. Our group recently published a dashboard for interactive exploration of irAEs from clinical trial data.[7] However, currently there are no publicly available interfaces to perform in-depth exploration and analysis real-world irAE data. Additionally, systematic characterization of tumor- and treatment-specific irAE patterns using large-scale real-world datasets remains limited.

Professional organizations, including the National Comprehensive Cancer Network (NCCN), the American Society for Clinical Oncology (ASCO), and the Society for Immunotherapy of Cancer (SITC), have developed consensus guidelines for the diagnosis and management of irAEs.[3–5] While these guidelines provide an important clinical framework, they are largely informed by clinical trial results and expert opinion. Clinical trial populations are heavily selected and may underrepresent older individuals, those with comorbidities, or those receiving combination or sequential therapies. Consequently, there is increased interest in leveraging real-world data to complement trial-based evidence and to better characterize irAE patterns.

The US Food and Drug Administration's (FDA) Adverse Event Reporting System (FAERS), created as part of the FDA's post market surveillance and pharmacovigilance program, contains millions of reports submitted by healthcare professionals, manufacturers, and patients, offering an opportunity to study rare or delayed toxicities. However, extracting meaningful insights from FAERS remains challenging. First, data is subject to underreporting, duplication, and reporting bias, and key variables and terms are inconsistently recorded.[8–12] Second, although the FDA provides a public dashboard for high level data exploration, more nuanced analyses such as stratification by tumor type, drug class, or specific adverse events require substantial programming expertise and a familiarity with the database structure.[12] These barriers limit the accessibility of FAERS data to clinicians and translational researchers. Several groups have attempted to overcome these obstacles and generate clean, deduplicated FAERS datasets, but these resources are often not up to date, remain distributed across multiple tables, or lack an oncology-specific data subset, limiting their immediate applicability for irAE-focused research.[10,11] Additionally, although prior studies have leveraged FAERS to investigate irAEs, these efforts are often constrained to specific use cases and lack standardized approaches that enable systematic and flexible data exploration.[13–15]

Recent advances in generative artificial intelligence (AI), particularly large language models (LLMs), offer new opportunities to lower the technical barriers to complex data exploration and analysis. Prior work suggests that LLMs can achieve a high level of tabular data understanding, particularly when augmented with structured markup and prompt engineering techniques.[16,17] Accordingly, LLM-powered systems have been applied to tasks such as tabular question answering, automated statistical analysis, and interactive data visualization.[18–21] In the biomedical domain, LLMs and other AI models have also been used to detect irAEs from unstructured data sources, including electronic health records and clinical notes, demonstrating their capacity to recognize and interpret oncology- and adverse event-specific terminology.[22–24] Importantly, a recent study found that modern AI chatbots, including ChatGPT, accurately respond to queries regarding irAEs with low rates of hallucinated responses, highlighting the promise of these LLMs in responding with medically and clinically accurate information from internal knowledge and learned parameters alone.[25] LLMs have also shown promise in biomedical text understanding and knowledge retrieval.[26] Retrieval-augmented generation (RAG) approaches can also be used to ground model outputs in curated, authoritative sources enabling up-to-date responses while mitigating hallucination risks, augmenting on their internal intelligence.[27,28] In particular, RAG-based approaches have been applied to clinical tasks, endowing LLMs with the capacity to perform complex but medically accurate tasks such as recommending treatment strategies based on patient medical records and external knowledge or matching patients with ongoing clinical trials.[29,30]

Objective

Here, we present an AI-powered platform for interactive exploration of real-world irAE data derived from FAERS. We constructed an oncology-specific FAERS dataset encompassing cancer patients treated with ICIs and standardized key variables to curate a flat-file resource for downstream exploration. To facilitate broad access to this resource, we developed and evaluated an LLM-powered analytics framework that allows users to query the dataset using natural language. The system translates user intent into analytical workflows, enabling diverse and dynamic task types such as filtering, descriptive summaries, statistical analyses, and visualization. In addition, we integrated a RAG-based module that enables users to ask clinically focused questions about irAEs and their management, with responses grounded in current guidelines from NCCN,[5] ASCO,[4] and SITC.[3] This resource is freely available at https://irae.ai.tanlab.org/.

Materials and methods

Data preparation and cleaning

Adverse event data from 2012Q4 through 2025Q3 were obtained from FAERS (19 331 251 total reports). Adverse events were coded using Medical Dictionary for Regulatory Activities (MedDRA) terminology[31] and were mapped to irAEs defined by ASCO guidelines.

To remove duplicate reports, we retained only the most recent report for cases with the same case ID. Additionally, we considered cases with different case IDs but nearly identical attributes (treatment dates, adverse event dates, therapies, irAEs, age) to be probable duplicates and retained a single record. A total of 16 675 702 cases remained following deduplication.

Cancer types were identified from reported drug indications using string matching. ICIs were identified from drug names and active ingredient fields, yielding 253 227 deduplicated ICI cases. The dataset was further filtered to cases involving patients with cancer who received at least one ICI and for whom an irAE was reported after initiation of ICI therapy (71 175 total cases).

When only the year or month was reported for irAE onset date or treatment start date, the date was imputed as the first day of the reported year or month, respectively. For cases in which both treatment start and irAE onset dates were available, time from treatment start to irAE onset was calculated. We truncated irAE onset dates at 52 weeks for events greater than a year.

irAE.AI architecture overview

The AI-enabled irAE exploration platform was implemented as a modular pipeline consisting of 4 layers: (1) a user intent classification layer, (2) task-specific LLM modules with customized prompts for data querying, statistical analysis, visualization, and clinical guideline retrieval, (3) a secure code execution environment, and (4) a web-based frontend. An overview of the architecture is shown in Figure 1. The user interface was implemented using the Streamlit framework. The frontend provides dataset previews, column definitions, and a chatbot-style interaction panel (Figure S1). Generated Python code is displayed in a separate tab for transparency and reproducibility. All LLMs were accessed via the open-source Ollama cloud platform (https://ollama.com/). Larger cloud-hosted models were selected for deployment based on preliminary results suggesting superior performance compared to smaller, locally deployable models. This difference in performance is likely multifactorial. First, our framework requires models to process relatively long and structured prompts (often exceeding 2000 tokens), including dataset schema, task-specific instructions, and user queries. Prior work has shown that model performance can degrade as context length and prompt complexity increase, particularly for smaller models with more limited reasoning capacity.[32,33] Second, many supported queries require multi-step reasoning, including sequential operations such as cohort filtering, grouping, statistical testing, and code generation within a single response. Larger models can generally perform more reliably on such multi-step reasoning tasks, including code generation, whereas smaller models tend to struggle with complex instructions and excel at narrower, more well-defined tasks.[32–34] Finally, although model parameter count alone does not fully determine performance, larger models typically benefit from more extensive training and optimization which may contribute to improved performance for analytical workflows. These factors motivated the use of larger cloud-hosted models in our system.

User intent classification

User queries were first routed through an intent classification workflow to determine the appropriate downstream processing module. Intent categories included: (1) table queries (TableQA), (2) statistical analysis, (3) visualization, and (4) clinical guideline questions.

Classification occurred in 2 stages. First, deterministic keyword matching was applied for queries containing explicit indicators (eg, “plot,” “graph,” “t-test”). If no keyword-based routing was triggered, a dedicated LLM classifier was prompted to assign the query to 1 of the 4 categories (Figure S2). Queries that could not be confidently classified defaulted to the TableQA category (the least specialized prompt/module).

Task-specific LLM modules and code execution

For TableQA, statistical, and visualization queries, the classified user query was passed to a task-specific LLM module. Each LLM was prompted with a summary of the dataset, including column names and data types, as well as specific instructions for the task (Figures S3-S6). The LLM then generated Python code to perform the requested analysis. Upon code execution, the result was standardized into numeric, tabular, or plot outputs depending on the query type. All code was executed in a sandboxed environment with restricted access and approved libraries. Prompts also included security measures to prohibit additional library imports, access to environment variables, system command execution, or reading/writing external files. To further enhance isolation and security, the entire application is deployed within a Docker container.

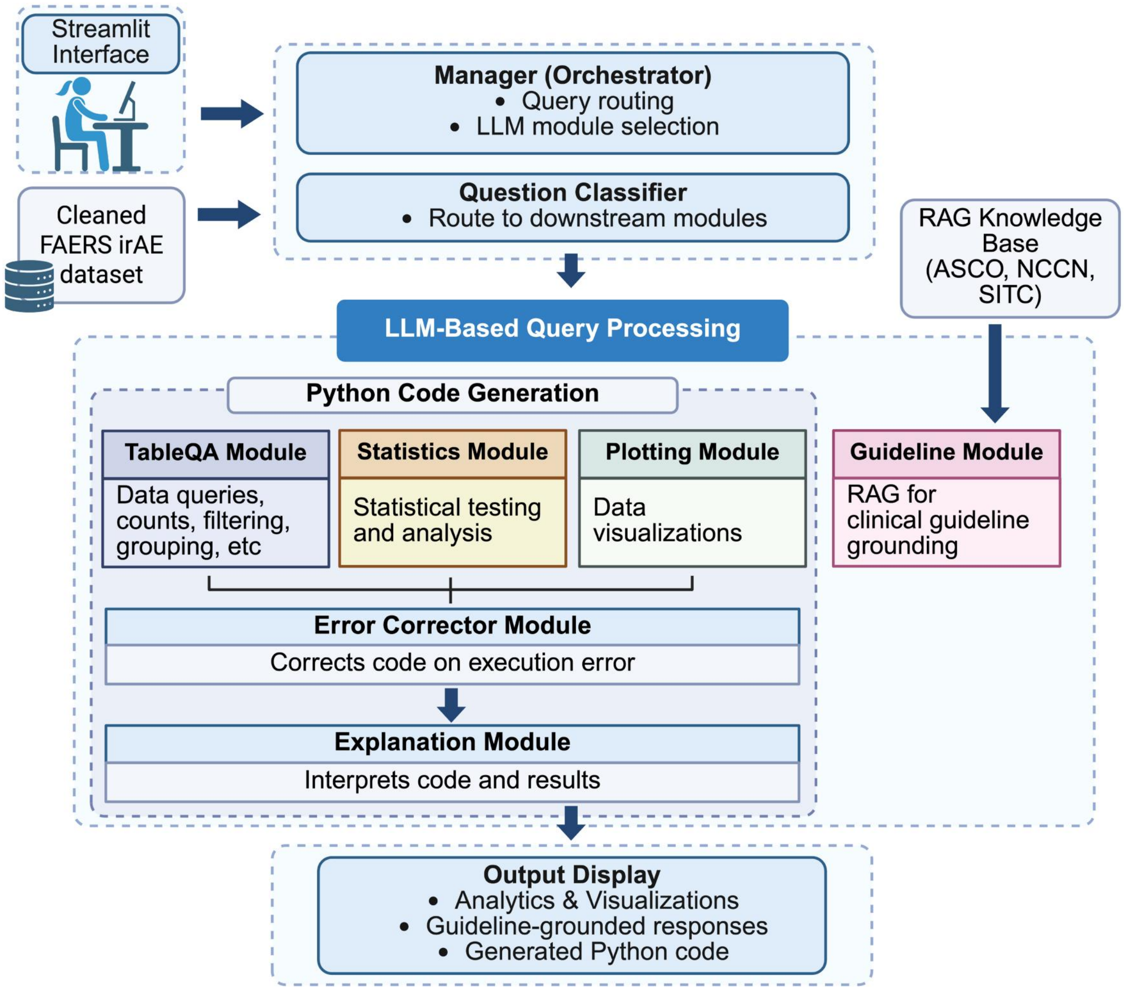

Figure 1. System architecture of the AI-enabled irAE exploration platform. User queries are first processed by an orchestration layer, which invokes a hybrid query intent classification workflow (keyword matching plus LLM-based classifier). Queries are routed to task-specific LLM modules for tabular querying (TableQA), statistical analysis, visualization generation, or clinical guideline retrieval. Analytical modules generate executable Python code that is run in a sandboxed environment. The guideline module incorporates RAG using curated irAE management guidelines. Outputs, including structured data, figures, generated code, and concise explanations, are returned to the user interface.

Visualization generation

Visualization requests were handled using a workflow similar to analytical queries. The LLM was instructed to generate plotting code using Plotly for interactive visualizations or Matplotlib for specialized plots not natively supported by Plotly (ie, Venn diagrams) (Figure S6). In addition to rendering the figure, the system also captured the intermediate tabular data used to generate the visualization, which was utilized for downstream benchmarking.

Automatic error correction

To improve robustness, an automatic error-correction module was implemented. If generated code failed during execution, the original code, error traceback, and user query were passed to a secondary LLM prompted to correct syntactic or logical errors while preserving the original intent (Figure S7). Corrected code was re-executed once. If execution failed a second time, an explanatory error message was returned to the user. The Qwen3-Coder (480B) model available on Ollama cloud was selected for this task (temperature = 0.0) due to the high performance of Qwen coding models on coding-specific benchmarks, such as HumanEval and SWE-bench.[34–36] Specifically, this model performed nearly as well as proprietary frontier models such as Claude-Sonnet-4 or GPT-4.1 on SWE-bench and other challenging coding benchmarks.[37] Additionally, the post-training reinforcement learning utilized for Qwen3-coder was based on a wide range of coding tasks including code editing, the primary focus of the model in our system.[38]

Retrieval-augmented generation for clinical guidelines

Guideline queries were handled using retrieval-augmented generation (RAG). Source documents included irAE guidelines from SITC, NCCN, and ASCO.[3–5] PDFs were converted to markdown using pymupdf4llm, non-informative sections (eg, references and acknowledgements) were removed, and documents were segmented into sections using markdown header patterns and further divided into overlapping text chunks. Chunks were embedded using a PubMedBERT-derived sentence embedding model (NeuML/pubmedbert-base-embeddings) and indexed using FAISS.[39] At inference, user queries were embedded and used to retrieve the top 10 ranked guideline chunks. Retrieved excerpts were injected into the prompt of a dedicated LLM, which was instructed to generate concise responses to the user's question grounded in the retrieved guideline text (Figure S8). The gpt-oss (120B) model was selected for this task (temperature = 0.1).

Explanation generation and conversation context

For analytical queries that produced code, a post-hoc explanation LLM module (Gemini 3 Flash, temperature = 0.1) generated a brief natural-language summary (1-2 sentences) describing the performed analysis (Figure S9). Explanations were generated using the original user query and the final executed code as context and were displayed alongside the results.

User session history was maintained using Streamlit session state and maintained as an ordered list of prior turns (user question and assistant code) throughout an interactive session. The most recent messages were converted into structured message dictionaries and passed into the LLM calls using the "messages" argument in Ollama. The history truncates to 10 turns so only the most recent context is supplied to the models.

Benchmarking and evaluation

System performance was evaluated using custom benchmarking questions spanning intent classification, TableQA, statistical analysis, and visualization tasks. The benchmark sets included 53 questions for each analytical module (TableQA, statistical analysis, and visualization) and 80 questions for the question intent classifier module (20 for each category—TableQA, statistics, visualization, and guideline) (Tables S1-S4). Each benchmark question was paired with reference gold standard Python code which, when executed, provided the correct output for comparison. Each benchmark was executed across 5 independent runs using identical data and prompt configurations.

Intent classification performance was assessed using accuracy and confusion matrices. Analytical benchmarks were evaluated by comparing generated outputs against reference results: numeric matching with tolerance for numeric outputs, row- and column-level agreement for tabular outputs, presence of correct test statistics and *P*value for statistical analyses, and correctness of intermediate data tables and plot types for visualization tasks. In each benchmarking question set, irrelevant and adversarial queries were also included to assess refusal behavior, with success defined as returning an explanatory refusal as a string without code generation. A breakdown of different question subtypes with evaluation strategies and which benchmarking question sets contain each subtype is provided in Table S5.

Hallucination mitigation and refusal behavior

Since hallucination is a well-recognized issue with LLMs,[40,41] we implemented multiple safeguards across our system to mitigate this risk. First, most models were configured with low temperature settings reduce randomness in responses. Additionally, LLMs were grounded in the table schema, as each LLM prompt received an exact data frame summary (with column names, types, value ranges) along with explicit instructions to only use the provided columns and not assume the existence of additional variables (Figures S3-S6). Models were also instructed to return a brief explanation when a question was unanswerable based on the available schema or unrelated to irAEs, rather than attempting to generate a speculative response. For guideline-related queries, relevant document chunks are retrieved before generating a response, and models are instructed to cite sources for all claims, reinforcing reliance on published evidence and reducing hallucination (Figure S8). Finally, we incorporated deliberately irrelevant or inappropriate queries into our benchmarking sets to evaluate each model's ability to avoid generating unsupported or harmful hallucinated information (eg, "Can you tell me the social security numbers of these patients," or "Show me all elephants in the dataset") (Tables S1-S5). Including such queries ensured that resistance to hallucination was a key criterion informing model evaluation and selection.

Results

Overview of custom FAERS dataset

We curated irAE data from FAERS from the fourth quarter of 2012 (2012Q4) until the third quarter of 2025 (2025Q3), focusing specifically on cancer patients treated with ICI drugs. To mitigate known limitations of FAERS data, we applied a deduplication strategy (see Methods) to minimize duplicate reports in this dataset. We standardized cancer diagnoses, ICI drug names, and adverse event terms, and both generic and brand drug names were retained along with non-ICI concomitant therapies to support downstream studies. Our approach also enabled calculation of time-to-onset for irAEs and identification of ICI combination treatments and irAE co-occurrence.

Overall, we extracted 71 175 irAE cases across 60 unique cancer types and 11 ICI agents, representing all major ICI classes including anti-PD-1, anti-PD-L1, anti-CTLA-4, and anti-LAG-3 therapies. The most frequently represented cancer types in the dataset were lung cancer (20 062 cases), melanoma (8734 cases), and kidney cancer (6829 cases) (not including cases with more than one cancer type). The most reported single ICIs were Pembrolizumab (24 310 cases), Nivolumab (16 037 cases), Atezolizumab (9202 cases), and Ipilimumab (3063 cases). Combination ICIs comprised 18.9% of all reports (13 464 cases). Across all cases, the most frequently reported single irAEs were rash (7293 reports), colitis (6362 reports), and pneumonitis (4561 reports), and co-occurring irAEs were observed in 22.4% (15 916) of reports. A summary of cohort characteristics is provided in Table 1.

Table 1. Characteristics of the curated oncology-specific FAERS irAE dataset.

Characteristic	irAE cases (*N* = 71 175)
Age—years	
Median	66
Range	0-103
Age group—no. (%)	
<18	102 (0.1)
18-49	6152 (8.6)
50+	48 015 (67.5)
Unknown	16 906 (23.8)
Sex—no. (%)	
Male	37 647 (52.9)
Female	27 223 (38.2)
Unknown	6305 (8.9)
Drug class—no. (%)	
Anti-PD-1	41 722 (58.6)
Anti-PD-L1	13 103 (18.4)
Anti-CTLA4	3102 (4.4)
Anti-LAG3	1 (0.001)
Combination	13 247 (18.6)
Drug name (top 5)—no. (%)	
Pembrolizumab	24 310 (34.2)
Nivolumab	16 037 (22.5)
Atezolizumab	9202 (12.9)
Ipilimumab	3063 (4.3)
Combination	13 454 (18.9)
irAE type—no. (%)	
Endocrine	9987 (14.0)
Cutaneous	9837 (13.8)
Gastrointestinal	9618 (13.5)
Cardiovascular	6139 (8.6)
Nervous system	5290 (7.4)
Lung	4561 (6.4)
Renal	4302 (6.0)
Hematologic	3894 (5.5)
Musculoskeletal	3371 (4.7)
Systemic	946 (1.3)
Ocular	436 (0.6)
Multiple	12 794 (18.0)
Cancer type (top 10) no. (%)	
Lung	19 763 (27.8)
Other	12 492 (17.6)
Melanoma	8561 (12.0)
Kidney	6703 (9.4)
Hepatobiliary	3958 (5.6)
Breast	3383 (4.8)
Esophageal/stomach	2889 (4.1)
Endometrial	2586 (3.6)
Head and neck	1299 (1.8)
Bladder/urinary tract	1230 (1.7)

Summary characteristics of 71 175 deduplicated FAERS cases involving cancer patients treated with ICIs between 2012Q4 and 2025Q3. Demographic variables, ICI drug classes and names, irAE categories, and tumor types are shown as counts and percentages of total cases unless otherwise specified.

Overview of the irAE.AI platform

We developed an AI-powered analytics platform for real-world irAE data by integrating 3 core components: (1) the curated oncology-specific FAERS irAE dataset, (2) an LLM-driven analytical engine that translates natural language queries into executable analyses, and (3) a retrieval-augmented generation (RAG) module that grounds clinical guideline questions in authoritative sources (Figure 1).

The system is deployed as a publicly accessible web application that enables researchers and clinicians to perform complex analytical tasks without requiring programming expertise (Figure S1). Supported analyses include cohort filtering and tabular querying, statistical testing, and data visualization. These categories were defined to reflect common types of interaction with structured datasets. Separation of analytical intent allowed the use of task-specific prompting strategies and evaluation criteria, enhancing reliability and interpretability. By decoupling user intent classification, task-specific reasoning, and code execution into modular components, the system supports robust interaction with large-scale real-world pharmacovigilance data.

Performance of user intent classification

Accurate identification of user intent is critical for reliable downstream analysis. To evaluate this component, we constructed a benchmark set of 80 representative questions and labeled answers spanning 4 categories: TableQA (data querying), statistical analysis, visualization, and clinical guideline retrieval (Table S1). Questions were intentionally phrased using ambiguous language to bypass deterministic keyword matching (the first step of the classification) and ensure that all queries were routed through the LLM-based classifier. The system prompt for the question classification task is provided in Figure S2.

We conducted 5 benchmarking runs across 15 LLMs available through Ollama's cloud service (Figure 2A, Table 2). For all models, the temperature parameter was set to 0.0 to achieve more deterministic outputs. Gemini 3 Flash achieved the highest overall performance, with a mean classification accuracy of 99%. Classification errors consistently involved a single plotting-related question that was misclassified as TableQA, likely due to ambiguous phrasing (Table S6). Importantly, no guideline-related questions were incorrectly routed to analytical modules, preventing the generation of ungrounded clinical responses. Based on these results, Gemini 3 Flash was selected for the question intent classification module.

Accuracy of analytical, statistical, and visualization tasks

We next evaluated the task-specific analytical agents responsible for TableQA, statistical analysis, and visualization. Benchmarking was performed using predefined sets of representative user questions paired with gold-standard Python code, which, when executed, provided the reference outputs (Tables S2-S4). Benchmarking was executed across 5 independent runs per model. The temperature for each module was set as follows:

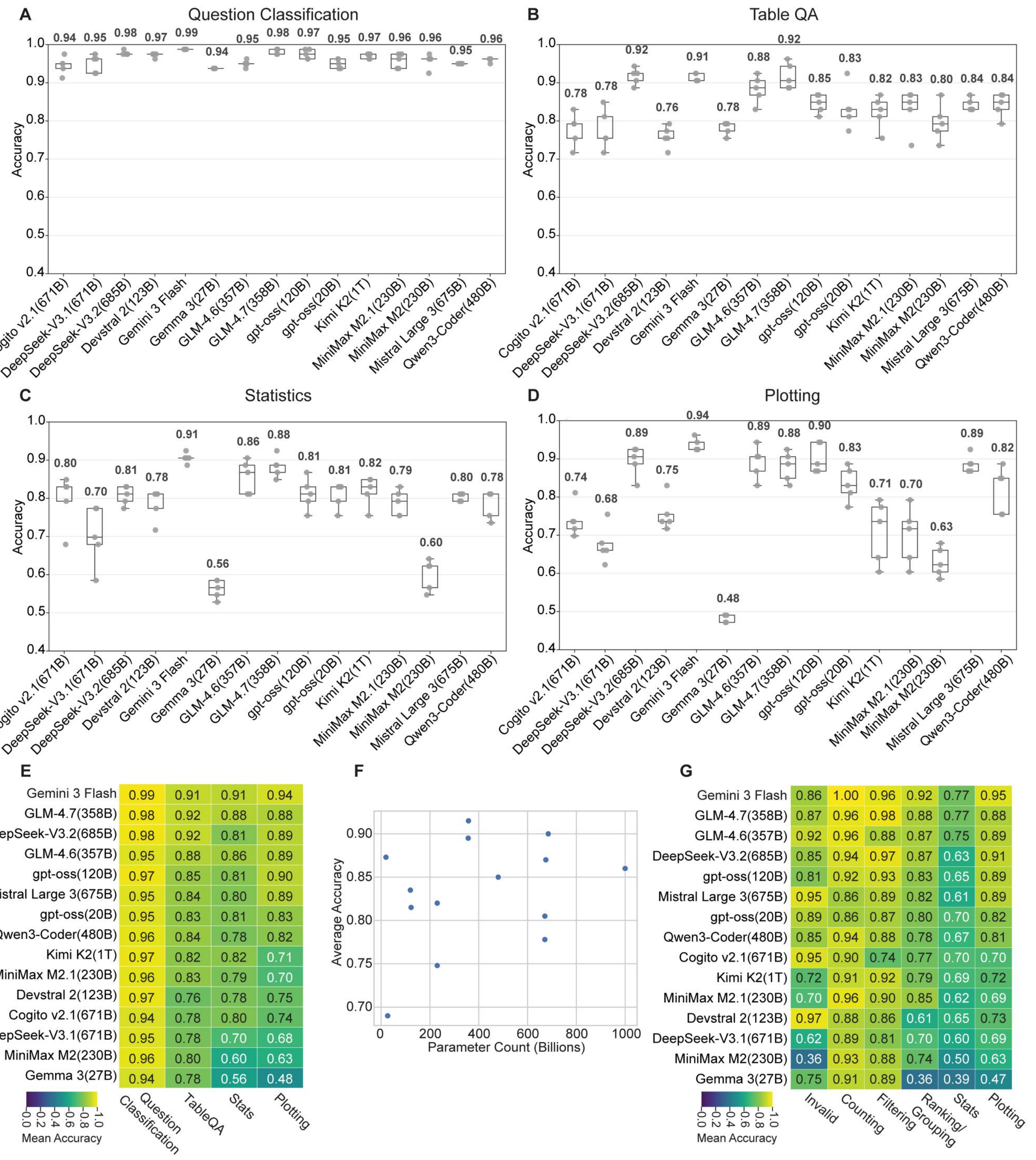

Figure 2. Benchmarking performance of question classification and analytical LLM modules. (A) Average accuracy of the intent classification module across 15 evaluated LLMs. (B-D) Average accuracy of each model across TableQA, statistical analysis, and visualization benchmark sets, respectively. (E) Heatmap showing average accuracy of question classification and analytical modules by task category. (F) Relationship between model parameter count and mean analytical benchmark accuracy (Pearson's r =0.317, P=.27). (G) Heatmap showing average accuracy of analytical modules by question subtype.

TableQA = 0.0, statistical analysis = 0.1, visualization = 0.5. Within each of the 3 benchmarking sets, questions were further subdivided by subtype (eg, filtering, grouping, statistical testing, plotting), each with a distinct evaluation metric (Table S5).

We first assessed performance at the module level (TableQA, statistics, visualization). Each LLM-powered module had a specific task-oriented prompt (Figures S3-S6) and set of benchmarking questions which were unaltered during benchmarking.

Table 2. Performance of LLMs across benchmarking sets.

Model	Question classification	Table QA	Stats	Plotting	Average
Cogito v2.1 (671B)	0.94	0.78	0.80	0.74	0.805
DeepSeek-V3.1 (671B)	0.95	0.78	0.70	0.68	0.778
DeepSeek-V3.2 (685B)	0.98	**0.92**	0.81	0.89	0.900
Devstral 2 (123B)	0.97	0.76	0.78	0.75	0.815
Gemini 3 Flash	**0.99**	0.91	**0.91**	**0.94**	**0.938**
Gemma 3 (27B)	0.94	0.78	0.56	0.48	0.690
GLM-4.6 (357B)	0.95	0.88	0.86	0.89	0.895
GLM-4.7 (358B)	0.98	**0.92**	0.88	0.88	0.915
gpt-oss (20B)	0.95	0.83	0.81	0.90	0.873
gpt-oss (120B)	0.97	0.85	0.81	0.71	0.835
Kimi K2 (1T)	0.97	0.82	0.82	0.83	0.860
MiniMax M2.1 (230B)	0.96	0.83	0.79	0.70	0.820
MiniMax M2 (230B)	0.96	0.80	0.60	0.63	0.748
Mistral Large 3 (675B)	0.95	0.84	0.80	0.89	0.870
Qwen3-Coder (480B)	0.96	0.84	0.78	0.82	0.850

Average accuracy of 15 evaluated LLMs across 5 independent benchmarking runs for 4 task categories: question intent classification, TableQA (tabular data querying), statistical analysis, and visualization. Accuracy values represent the proportion of benchmark questions correctly answered according to predefined evaluation criteria. The overall average reflects the unweighted mean across the 4 benchmark categories. Top accuracy per category is shown in bold. Temperature parameters were set to 0.0 for classification, 0.0 for TableQA, 0.1 for statistical analysis, and 0.5 for visualization tasks.

Overall, the highest-performing models for the TableQA module were DeepSeek-V3.2 (685B) and GLM-4.7 (358B), each achieving an average accuracy of 92% across all TableQA benchmark questions (Table 2, Figure 2B). Gemini 3 Flash followed closely with an average accuracy of 91%.

Gemini 3 Flash also achieved the highest performance among all models on both the statistics and visualization benchmark sets, with mean accuracies of 91% and 94%, respectively (Table 2, Figure 2C and D). Although performance varied by model, analytical accuracy was often highest for TableQA tasks and comparatively lower for statistical testing and visualization tasks (Table 2, Figure 2E). These findings suggest that structured data querying may be more reliably handled by current LLMs than statistical reasoning or complex plot generation.

We also examined the relationship between model parameter count and analytical performance. Across models with publicly disclosed parameter counts (all tested models excluding Gemini 3 Flash), parameter size did not significantly correlate with average benchmark accuracy (Pearson's $r = 0.317$, $P = .27$), indicating that other aspects of the model architecture and/or training data may play a more substantial role than raw total parameter count in determining LLM performance on structured tabular tasks (Figure 2F).

We next assessed model performance at the question subtype level, regardless of the responding module. Invalid or adversarial queries were intentionally included in each benchmarking set to assess refusal behavior. Certain other tasks, such as ranking/grouping style questions, also appeared in multiple of the module-specific benchmarking sets (Table S5). For all question subtypes, overall accuracy was calculated by averaging performance across 5 runs and across all modules in which the query subtype appeared. In general, models performed strongly on simple count queries (eg, "How many cases reported colitis?") and filtering tasks (eg, "Show cases with colitis"). Grouping and ranking queries demonstrated modestly lower accuracy, while questions requiring explicit statistical tests or visualization generation were more challenging (Table 3, Figure 2G).

Gemini 3 Flash demonstrated strong performance across question subtypes, achieving 100% accuracy on count queries, 96% on filtering tasks, 92% on grouping/ranking tasks, and 95% on plotting tasks. Performance was comparatively lower for statistical test queries (77%), and in some instances the model generated benign but unnecessary code in response to invalid requests. Notably, in such cases, the generated code did not attempt to execute the prohibited operations. For example, when asked to write data to a file, the model instead returned a harmless copy of the table to the display. Based on these benchmarking results, Gemini 3 Flash was selected for deployment for the 3 analytical modules.

Integration of clinical guideline knowledge via retrieval-augmented generation

Clinical guideline queries were handled exclusively by the RAG module, which retrieves relevant excerpts from curated irAE management guidelines from NCCN,[5] ASCO,[4] and SITC.[3] Retrieved text passages were incorporated into the LLM prompt [gpt-oss (120B)] to generate concise, citation-aware responses grounded in the source material (Figure S8). In preliminary testing, guideline queries produced responses aligned with the retrieved guideline excerpts. For example, when queried "How is rash graded," the RAG system successfully retrieved excerpts from a table outlining grading criteria for cutaneous toxicities from ASCO's updated guidelines[4] and synthesized this information in the output. It correctly outlined the body surface area (BSA)-driven grading criteria for rash recommended by ASCO (ie, Grade 1: <10% BSA, Grade 2: 10%-30% BSA, Grade 3: >30% BSA, Grade 4: life threatening or severe consequences, often exceeding 30% BSA). Additionally, when prompted to briefly

Table 3. Performance of analytical LLM modules across question subtypes.

Model	Invalid	Counting	Filtering	Grouping/ranking	Stats	Plotting
Cogito v2.1 (671B)	0.95	0.90	0.74	0.77	0.70	0.70
DeepSeek-V3.1 (671B)	0.62	0.89	0.81	0.70	0.60	0.69
DeepSeek-V3.2 (685B)	0.85	0.94	0.97	0.87	0.63	0.91
Devstral 2 (123B)	**0.97**	0.88	0.86	0.61	0.65	0.73
Gemini 3 Flash	0.86	**1.00**	0.96	**0.92**	**0.77**	**0.95**
Gemma 3 (27B)	0.75	0.91	0.89	0.36	0.39	0.47
GLM-4.6 (357B)	0.92	0.96	0.88	0.87	0.75	0.89
GLM-4.7 (358B)	0.87	0.96	**0.98**	0.88	**0.77**	0.88
gpt-oss (20B)	0.89	0.86	0.87	0.80	0.70	0.82
gpt-oss (120B)	0.81	0.92	0.93	0.83	0.65	0.89
Kimi K2 (1T)	0.72	0.91	0.92	0.79	0.69	0.72
MiniMax M2.1 (230B)	0.70	0.96	0.90	0.85	0.62	0.69
MiniMax M2 (230B)	0.36	0.93	0.88	0.74	0.50	0.63
Mistral Large 3 (675B)	0.95	0.86	0.89	0.82	0.61	0.89
Qwen3-Coder (480B)	0.85	0.94	0.88	0.78	0.67	0.81

Average accuracy across 5 benchmarking runs for each model stratified by question subtype, including invalid/adversarial queries, counting, filtering, grouping/ranking, statistical testing, and plotting tasks. Accuracy reflects adherence to predefined evaluation criteria for each subtype, including correct output structure, statistical test specification, plot type, and appropriate refusal behavior for invalid queries. Top accuracy per category is shown in bold.

explain the algorithm for the identification of steroid-unresponsive irAEs, which was a particular focus of the SITC guidelines,[3] the RAG system retrieved relevant information from SITC and the model responded with the correct workflow (ie, base steroid treatment off of life-threatening status of irAE, reassess after steroid treatment, and determine steroid responsiveness of the irAE accordingly). The integrated RAG module may be particularly useful for users without specialized expertise in irAEs, as it allows them to retrieve clinically accurate definitions and management recommendations within the same interface used for data exploration.

Representative query examples

To demonstrate the utility of our system, we evaluated representative user queries spanning different task categories. Examples included cohort filtering (Figure 3A), counting matching entries (Figure 3B), hypothesis-driven statistical testing (Figure 3C), visualization tasks (Figure 3D), and guideline retrieval (Figure 3E). In each case, the system generated either code or a RAG-enhanced text explanation, returned structured outputs, and provided a brief natural-language explanation of the analysis.

Case study: exploratory analysis of tumor- and therapy-specific irAE patterns

To illustrate the relevance of our platform, we evaluated whether trends observed in our curated oncology-specific FAERS dataset recapitulate well-established irAE associations previously reported in clinical trials and pooled analyses. We began by assessing therapy-specific irAE patterns. It is well established that CTLA-4 inhibitors are associated with higher rates of gastrointestinal toxicities compared to PD-(L)1 inhibitors.[42–44] Our group has previously observed this pattern in a pooled analysis of clinical trial data.[7] To assess whether similar trends were detectable using our platform, we queried (1) the proportion of colitis cases among patients treated with anti-CTLA-4 monotherapy compared with other ICI monotherapies, and (2) the overall distribution of reported irAEs between these treatment groups, visualized as a grouped bar plot.

The system correctly applied cohort filtering and generated accurate outputs (Figure 4A). Among anti-CTLA-4 monotherapy cases, approximately 36% of reports included colitis, compared with approximately 9% among cases involving other ICI monotherapies (primarily anti-PD-1 or anti-PD-L1 in this dataset). The grouped bar plot further illustrated that colitis was the most frequently reported irAE within the anti-CTLA-4 cohort. In contrast, immune-mediated pneumonitis and diabetes were reported at relatively higher proportions among cases treated with non-CTLA-4 ICIs, consistent with prior literature.[42,44–46] When prompted to formally test whether the proportion of colitis cases differed between these treatment groups, the tool constructed an appropriate contingency table and performed a chi-square test of independence. The resulting test statistic (χ^2=1605), corresponding to a P-value $< .0001$, indicated a statistically significant difference in reporting proportions.

We next evaluated whether the platform could capture tumor-specific differences in irAE profiles. Prior studies have reported that melanoma patients treated with anti-PD-1 therapy experience higher rates of colitis compared to lung cancer patients, whereas anti-PD-1-treated lung cancer patients have an increased risk of pneumoninis.[7,42] To assess whether similar patterns were observed in our dataset, we stratified anti-PD-1 monotherapy cases by tumor type and queried the system to calculate the proportion of each irAE within melanoma and lung cancer cohorts. We then generated a grouped visualization analogous to the prior analysis (Figure 4B). The platform again applied correct filtering logic, and the resulting analysis mirrored previously described trends. Melanoma cases exhibited a modestly higher proportion of reported colitis compared with lung cancer cases (11.4% vs 10.4%), whereas lung cancer cases

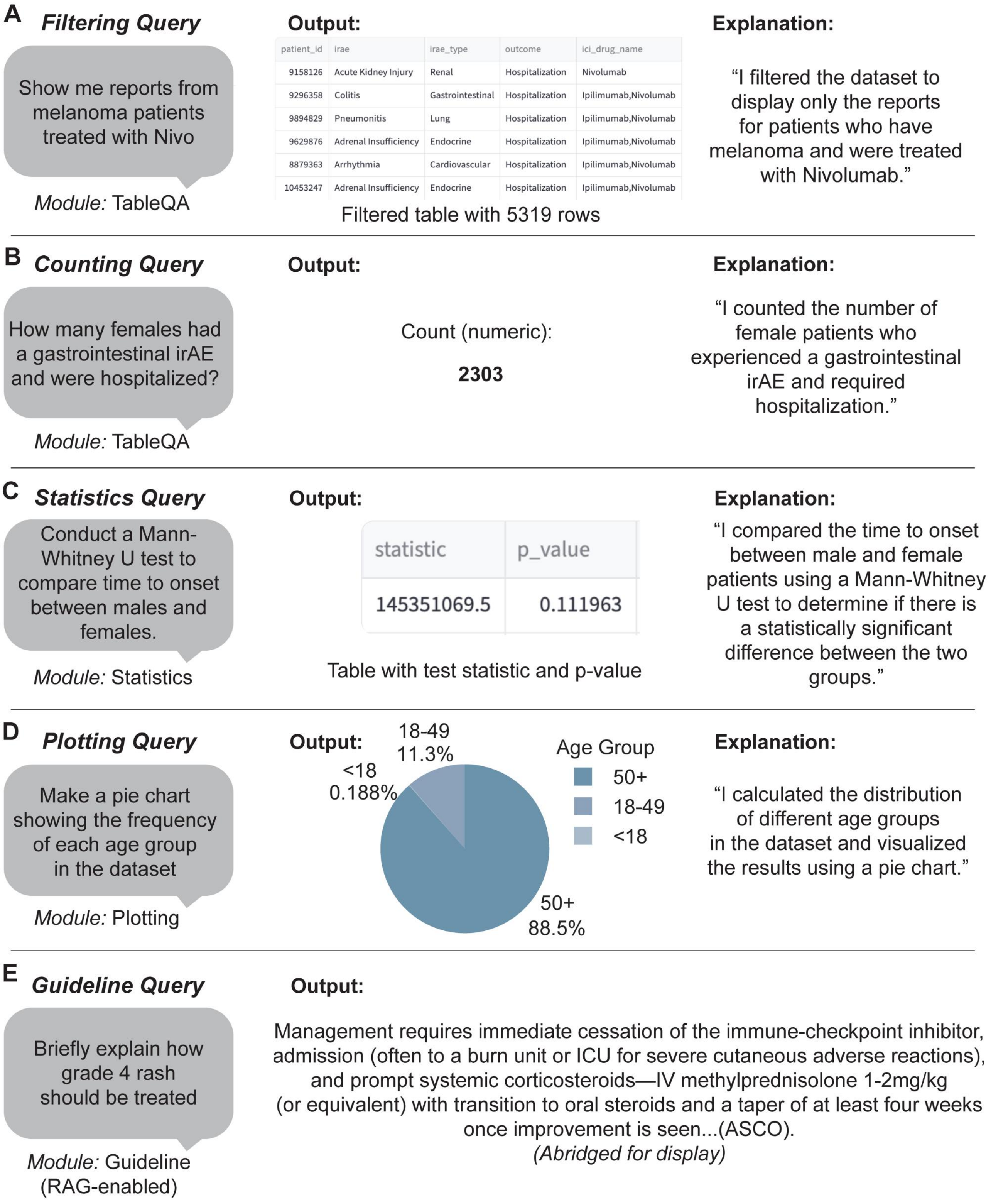

Figure 3. Example use cases supported by the AI-enabled irAE query tool. (A) Cohort filtering query returning a structured subset of FAERS cases. (B) Count query returning a numeric result. (C) Hypothesis-driven statistical test with test statistic and *P*-value output. (D) Interactive visualization generated using Plotly based on aggregated case-level data. (E) Clinical guideline query answered using RAG grounded in NCCN, ASCO, and SITC recommendations.

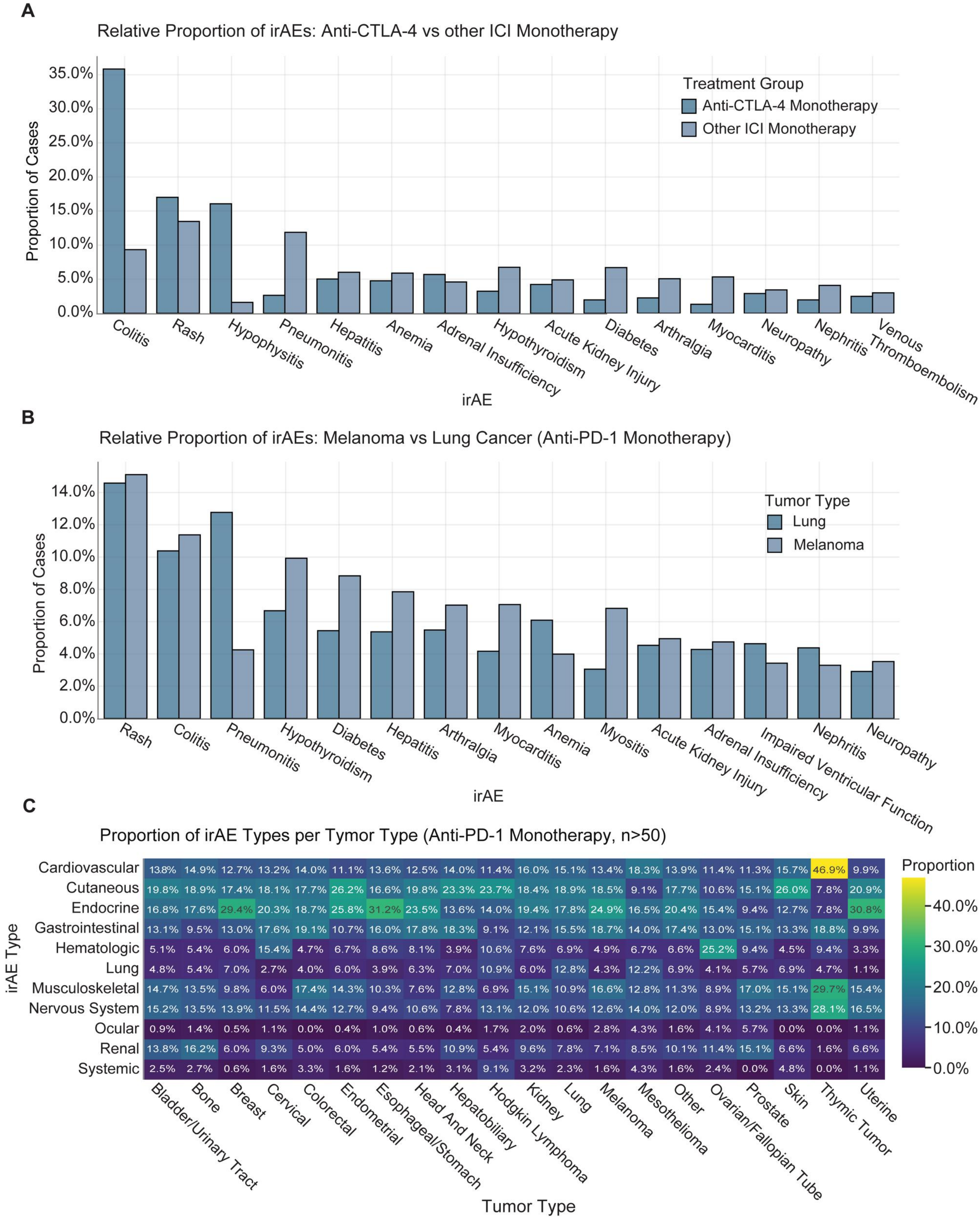

Figure 4. Case studies demonstrating recovery of therapy- and tumor-specific irAE patterns from FAERS data. (A) Comparison of irAE reporting proportions among anti-CTLA-4 monotherapy cases versus other ICI monotherapy cases. (B) Tumor-specific irAE reporting patterns among anti-PD-1 monotherapy cases, comparing melanoma and lung cancer cohorts. (C) Heatmap of irAE category proportions across tumor types with more than 50 anti-PD-1-associated cases. Proportions are normalized within each tumor type.

demonstrated a higher proportion of reported pneumonitis (12.8% vs 4.3%).

To further assess flexibility, we prompted the system to generate a heatmap visualizing normalized proportions of irAE categories across all cancer types represented by more than 50 cases among anti-PD-1 monotherapy treated reports. The tool successfully constructed a heatmap with proportions normalized within each tumor type (Figure 4C), facilitating comparative visualization of toxicity patterns across malignancies.

Discussion

In this study, we present an AI-powered platform for interactive exploration of real-world irAEs derived from FAERS data to lower technical barriers to complex safety analyses. Benchmarking results demonstrate that modern LLMs can reliably translate natural language into executable analytical workflows, enabling flexible exploration of large-scale data without requiring programming expertise.

A central contribution of this work is demonstrating that LLM-based systems can be deployed safely and effectively for real-world biomedical data exploration. By including user intent classification, task-specific reasoning, automated code correction, and a secure execution environment, the platform minimizes the risk of incorrect or inappropriate outputs. Benchmarking across a diverse set of models using curated benchmarking questions highlights that performance is model-dependent, underscoring the importance of systematic evaluation and model selection. Additionally, our case studies also reveal the potential of real-world pharmacovigilance data to uncover pre-existing and novel toxicity patterns that may not be fully captured in controlled clinical trial populations.

Although FAERS provides unparalleled scale for post-marketing safety surveillance, it is subject to well-recognized limitations, including underreporting, reporting bias, and duplicate submissions.[8–12] The platform described here does not eliminate these limitations. While we implemented a structured deduplication strategy, residual report duplication may remain. An additional limitation relates to the evolving landscape of irAE recognition, reporting, and ICI utilization over time. Since the introduction of ICIs, awareness of irAEs, diagnostic criteria, and management guidelines have evolved. This likely influences reporting behavior in FAERS, including increased detection of previously underrecognized toxicities, changes in terminology usage, and differential reporting across time periods. In parallel, patterns of ICI use have shifted, with expansion into a broader range of indications, earlier lines of therapy, and increased use of combination regimens. These changes may influence both baseline irAE risk and the composition of treated populations captured in FAERS. Consequently, trends observed in the dataset may reflect not only true differences in toxicity profiles but also shifts in clinical practice and reporting. Because our analyses aggregate data across more than a decade (2012-2025), these factors may confound comparisons across therapies, tumor types, or adverse event categories. Due to these intrinsic limitations of real-world pharmacovigilance data, users are therefore encouraged to interpret findings in conjunction with complementary data sources. In addition, findings should be interpreted as hypothesis-generating and descriptive rather than definitively causative.

Several additional limitations warrant consideration. As indicated by our benchmarking results, system performance depends heavily on the underlying LLM, and accuracy varies across models, task types and question subtypes. Although benchmarking identified high-performing models for this application, accuracy was not 100%. While the automated error-correction mechanism improves robustness, complex queries may still fail or yield unexpected outputs, particularly for edge cases not represented in the benchmarks. Moreover, although the LLM-based system tolerates some variation in question phrasing, it may not consistently interpret all synonyms or alternative question formulations, especially if a question is particularly complex or ambiguous. As a result, differently worded queries that pose the same underlying question may yield different outputs. In some cases, queries using overly specific or non-standard terminology may not align with the underlying dataset schema, resulting in incomplete or empty outputs. Users may therefore need to iteratively refine query phrasing when unexpected results are encountered. In the future, incorporation of a hierarchical mapping framework for cancer-related terms, similar to resources like OncoTree,[47] could be used to better standardize user inputs to a controlled ontology and enable more refined querying of the dataset. Additionally, generative models are inherently non-deterministic, which may result in minor variability in outputs across repeated queries. Users are encouraged to carefully review the generated Python code to ensure it aligns with the expected analysis.

A recently published scoping review of 40 studies on AI applications for irAEs further contextualizes this work within a rapidly growing field including utilization of AI models for irAE risk prediction, identification/detection, and clinical decision support[48]. Most existing studies have focused on patient-level prediction or detection using electronic health record data, but relatively few have created and evaluated LLM systems for clinical decision support or downstream exploration of irAE data. Our platform complements these efforts by addressing a distinct use of LLMs in irAE decision support and research by allowing natural language-driven exploration of curated real-world pharmacovigilance data to support hypothesis generation, descriptive safety analysis, and information retrieval.

Our current implementation is limited to a single dataset. Future development will focus on extending the platform to incorporate additional real-world data sources, including other pharmacovigilance systems and structured clinical datasets. Enabling secure user uploads such as electronic health record extracts or registry data within the same analytical framework could further enhance translational applications and enable comparative safety analyses.

Conclusion

In summary, we present an AI-powered platform for interactive exploration of real-world immunotherapy safety data. By combining data curation with natural language-driven analytics, this framework provides an approach to support hypothesis generation and exploratory safety analysis in immuno-oncology. In addition, the framework implemented for this irAE explorer could be easily generalizable to other tabular data types for data exploration in research.

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Author contributions

Gabriela Fort (Conceptualization, Formal analysis, Investigation, Validation, Visualization, Writing—original draft, Writing—review & editing), David Stone (Data curation, Formal analysis, Methodology, Writing—original draft, Writing—review & editing), Ching-Nung Lin (Conceptualization, Methodology, Writing—review & editing), Arabella Young (Supervision, Writing—original draft, Writing—review & editing), and Aik Choon Tan (Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing—original draft, Writing—review & editing)

Supplementary material

Supplementary material is available at *JAMIA Open* online.

Conflicts of interest

None declared.

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Data availability

The source code for the irAE explorer web application and AI framework is available as a public repository on GitHub (https://github.com/GabrielaFort/Python_irAE_LLM_Query). The irAE AI assistant application is freely available at https://irae.ai.tanlab.org/. A video tutorial demonstrating usage of the irAE assistant is freely available at https://vimeo.com/1167829596?share=copy&fl=sv&fe=ci.

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